

AITIR Technical Exhibit

Framework Architecture and Operational Model

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Proposed Endeavor: AI-Assisted Adaptive Threat Intelligence and Response for Public-Sector and Law-Enforcement Cybersecurity

1. Purpose of This Exhibit

This exhibit explains the technical and operational model of the AI-Assisted Adaptive Threat Intelligence and Response Framework, known as AITIR. It is intended to help USCIS understand the specific cybersecurity undertaking at the center of Mr. Hossain's National Interest Waiver petition.

AITIR is not a general claim that artificial intelligence is useful in cybersecurity. It is a public-sector cybersecurity framework focused on identity-centered threat detection, adaptive threat intelligence, risk prioritization, and response support for government and law-enforcement environments.

2. Problem: Identity and Access Threats in Public-Sector Systems

Public-sector organizations rely on identity and access systems to control who can reach sensitive systems, criminal justice information, government applications, remote-access services, cloud platforms, endpoints, and shared information environments. These systems are essential, but they also create a major attack path.

Many cybersecurity incidents begin or escalate through identity-related activity: credential theft, unauthorized access, abnormal login behavior, privileged-account misuse, improper access changes, compromised remote access, or insider activity. In law-enforcement and government environments, these threats are especially serious because systems often contain sensitive investigative, public-safety, citizen-service, operational, or regulated information.

- Compromised credentials can allow an attacker to appear as an authorized user.
- Improper privilege escalation can expose sensitive records or mission-critical applications.
- Delayed prioritization of access anomalies can slow incident response.
- Poorly contextualized alerts can overwhelm security teams and bury the most important risks.

The public-sector consequence is larger than an internal IT problem. Identity and access failures can affect public services, law-enforcement operations, information sharing, auditability, continuity, and public trust.

3. Current Gap: Static IAM and Periodic Access Reviews Are Not Enough

Identity and access management programs commonly include account provisioning, deprovisioning, password or authentication controls, role assignments, access approvals, and periodic access reviews. These controls are necessary. They do not fully solve the adaptive-threat problem.

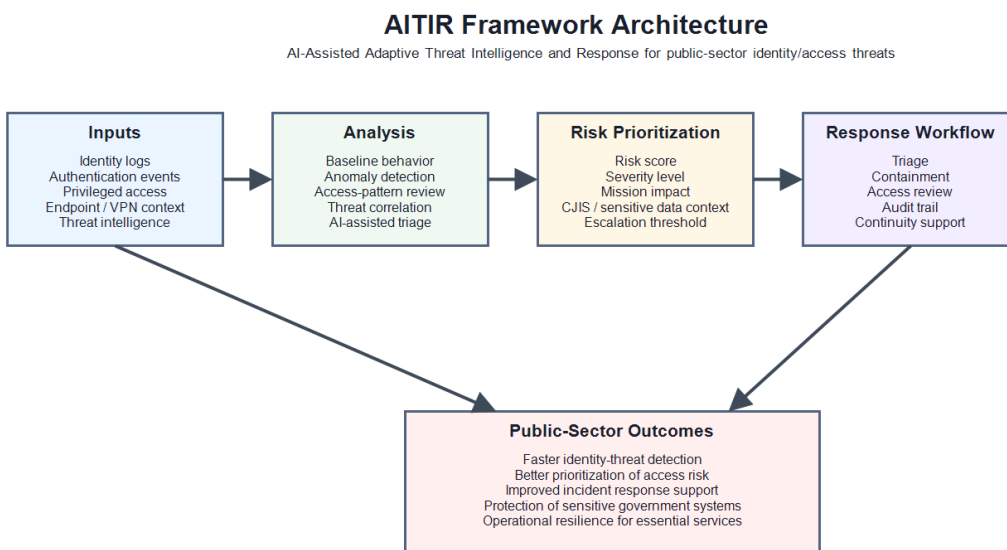
Static IAM reviews often answer the question, "Who has access?" AITIR is designed to help answer a more operational question: "Which identity and access activity indicates elevated cyber risk right now, and what response should receive priority?"

- Periodic access reviews may miss abnormal behavior between review cycles.
- Role-based access controls may not detect account misuse after credentials are compromised.
- Authentication events may be reviewed separately from endpoint, VPN, or threat-intelligence context.
- Security teams may receive too many alerts without a clear risk-prioritization workflow.
- Incident response may be delayed when identity, access, endpoint, and operational context are not evaluated together.

This gap creates the need for a framework that treats identity and access activity as an active threat-intelligence source and connects it to response prioritization.

4. AITIR Model Overview

AITIR integrates five operational layers: inputs, analysis, risk prioritization, response workflow, and public-sector outcomes.



This diagram is a generalized framework model and does not disclose confidential employer systems or sensitive operational details.

Figure 1. Generalized AITIR Framework Architecture. The diagram is a non-confidential model and does not disclose employer systems, sensitive law-enforcement information, or protected operational details.

4.1 Inputs

AITIR begins with security-relevant signals from public-sector technology environments. The framework is designed to combine identity, access, endpoint, remote-access, and threat-intelligence context.

- Identity and access logs, including user accounts, role assignments, privilege changes, and access approvals.
- Authentication events, including successful and failed logins, unusual timing, location anomalies, and multi-factor activity.

- Privileged-account behavior, including administrative access, sensitive-system access, and abnormal permission use.
- Endpoint, VPN, and remote-access context that helps determine whether access activity is normal or suspicious.
- Threat-intelligence indicators relevant to credential misuse, phishing, malware, ransomware, account takeover, or abnormal access patterns.

4.2 Analysis

The analysis layer evaluates identity and access events in context. It does not treat every access event as equally important. It looks for patterns that suggest elevated risk.

- Baseline behavior comparison: identifying activity that differs from normal user or role behavior.
- Anomaly detection: flagging abnormal access timing, location, volume, privilege use, or system access.
- Threat correlation: connecting identity activity with endpoint, VPN, remote-access, or known threat indicators.
- AI-assisted triage: helping classify which events require immediate review and which are lower priority.

4.3 Risk Prioritization

The risk-prioritization layer ranks events based on severity, sensitivity, mission impact, and response urgency. This is important because public-sector security teams must decide what to investigate first.

- Risk score: an assigned level based on behavior, context, and potential impact.
- Sensitive-system context: whether the access involves criminal justice information, public services, regulated information, or mission-critical applications.
- Operational impact: whether the event could affect continuity, investigation, public service, or agency coordination.
- Escalation threshold: when an event should move from monitoring to triage, containment, audit, or incident response.

4.4 Response Workflow

The response workflow connects prioritized identity risk to action. AITIR is intended to support, not replace, human cybersecurity judgment.

- Triage: determine whether the event is benign, suspicious, or urgent.
- Containment: restrict, reset, or review access where risk is elevated.
- Access review: verify role, permission, and account legitimacy.
- Audit trail: preserve evidence of access activity, decision-making, and response action.
- Continuity support: reduce risk while maintaining essential public-sector operations where possible.

5. Public-Sector Use Cases

5.1 Law Enforcement Systems

AITIR is directly relevant to law-enforcement technology environments because these systems often support criminal investigations, digital evidence, intelligence sharing, criminal justice information, public safety operations, and secure interagency communication.

- Detect abnormal access to sensitive investigative or criminal justice systems.
- Prioritize privileged-account activity that may affect evidence, records, or secure information sharing.
- Support faster review of suspicious authentication or remote-access events.
- Preserve auditability while supporting operational continuity.

5.2 Government Services

Public agencies deliver services through portals, databases, case-management systems, payment systems, health or emergency-service platforms, and internal workflows. Identity compromise or access misuse can disrupt service delivery and expose sensitive citizen information.

- Identify access anomalies affecting citizen-facing systems.
- Prioritize threats involving sensitive public records or service platforms.
- Help agencies maintain continuity while investigating access-related risks.

5.3 Public-Sector Infrastructure

State and local governments, education institutions, public health entities, and critical public-service organizations depend on hybrid networks, endpoints, cloud platforms, and remote-access tools. AITIR provides a model for connecting access activity to broader cyber resilience.

- Correlate remote-access behavior with endpoint or network context.
- Improve prioritization of identity-related alerts in resource-constrained environments.
- Support incident-response decisions when access activity indicates elevated risk.

6. Relationship to NIST and Public-Sector Security Practice

AITIR aligns with major security-control concepts commonly used in government and public-sector environments. It is designed to support, not replace, existing cybersecurity frameworks.

- NIST SP 800-53: access control, identification and authentication, audit and accountability, incident response, risk assessment, system monitoring, and contingency planning.
- NIST Cybersecurity Framework 2.0: govern, identify, protect, detect, respond, and recover functions.
- Digital identity and zero trust concepts: continuous assessment of identity, access, device, and contextual risk.
- Incident-response practice: triage, containment, evidence preservation, remediation, and continuity.

This alignment shows that AITIR is not a disconnected academic concept. It is a practical framework intended to improve how established public-sector security controls can be operationalized around identity-centered threat detection and response.

7. Future Development Roadmap

AITIR will continue to be developed through a phased roadmap that supports both research dissemination and practical public-sector review.

Phase I: Framework Refinement

- Finalize AITIR architecture, terminology, and threat-to-response workflow.
- Define identity-threat indicator categories and risk-prioritization logic.
- Prepare a non-confidential implementation narrative suitable for public-sector review.

Phase II: Research and Publication

- Prepare additional manuscripts on AI-assisted identity-threat detection, public-sector cyber resilience, and adaptive response models.
- Connect AITIR concepts to NIST SP 800-53, NIST CSF 2.0, zero trust, and incident-response practice.

Phase III: Practitioner Dissemination

- Publish public cybersecurity articles explaining identity-centered risk, access misuse, and adaptive threat intelligence.
- Share non-confidential AITIR materials with cybersecurity professionals, independent experts, and public-sector security practitioners for feedback.

Phase IV: Framework/Model Release

- Release a generalized AITIR framework overview, architecture model, and operational workflow.
- Continue refining the model through publications, professional service, peer review, and practitioner feedback.

8. Why This Technical Exhibit Supports the NIW Petition

This technical exhibit supports Prong 1 because it shows that the proposed endeavor addresses a concrete national cybersecurity problem: identity-centered threats in public-sector systems. It supports Prong 2 because it shows that Mr. Hossain has a specific, defined framework already tied to his publications, public-sector experience, KBI work, and expert letters. It supports Prong 3 because the framework is portable beyond one employer and can continue through documentation, publication, professional dissemination, and public-sector outreach.

The exhibit therefore helps distinguish the proposed endeavor from routine cybersecurity employment. AITIR is a named framework with a defined problem, technical model, public-sector use cases, and future development path.